

Lab 02 DNS

All screenshot provided in each 'Question'

1. Nslookup

1. Run nslookup to obtain the IP address of a Web server in Asia. What is the IP address of that server?

Here are multiple addresses shown:

108.138.94.105
108.138.94.127
108.138.94.54
108.138.94.74

```
> nslookup www.nifty.com

Server:         192.168.0.1
Address:        192.168.0.1#53

Non-authoritative answer:
Name:   www.nifty.com
Address: 108.138.94.105
Name:   www.nifty.com
Address: 108.138.94.127
Name:   www.nifty.com
Address: 108.138.94.54
Name:   www.nifty.com
Address: 108.138.94.74
```

2. Run nslookup to determine the authoritative DNS servers for a university in Europe.

I ran the nslookup to determine the authoritative DNS servers for Oxford University is ~~primary.dns.ox.ac.uk~~ as below.

dns.ox.ac.uk

```
> nslookup -type=NS ox.ac.uk dns0.ox.ac.uk

Server:         dns0.ox.ac.uk
Address:        129.67.1.190#53

ox.ac.uk       nameserver = auth5.dns.ox.ac.uk.
ox.ac.uk       nameserver = auth6.dns.ox.ac.uk.
ox.ac.uk       nameserver = dns0.ox.ac.uk.
ox.ac.uk       nameserver = dns2.ox.ac.uk.
ox.ac.uk       nameserver = auth4.dns.ox.ac.uk.
ox.ac.uk       nameserver = dns1.ox.ac.uk.
```

3. Run nslookup so that one of the DNS servers obtained in Question 2 is queried for the mail servers for Yahoo! mail. What is its IP address?

98.136.96.76

98.136.96.75

67.195.204.79

67.195.228.94

98.136.96.77

67.195.228.111

67.195.228.110

67.195.228.106

Note: Oxford's DNS refuses external queries, so I use public resolver

```
> nslookup -type=MX yahoo.com 8.8.8.8

Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
yahoo.com    mail exchanger = 1 mta7.am0.yahoodns.net.
yahoo.com    mail exchanger = 1 mta6.am0.yahoodns.net.
yahoo.com    mail exchanger = 1 mta5.am0.yahoodns.net.

Authoritative answers can be found from:

> nslookup mta5.am0.yahoodns.net 8.8.8.8

Server:      8.8.8.8
Address:     8.8.8.8#53

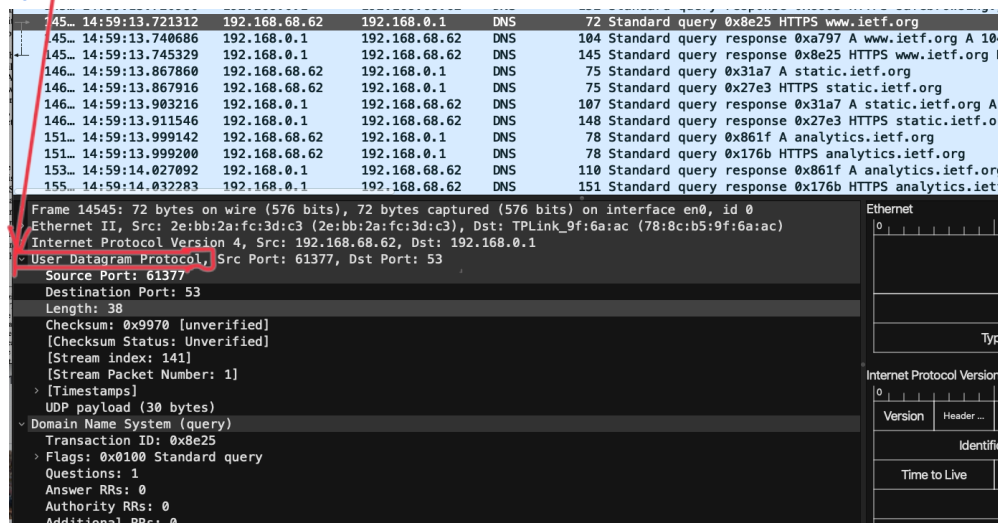
Non-authoritative answer:
Name:   mta5.am0.yahoodns.net
Address: 98.136.96.76
Name:   mta5.am0.yahoodns.net
Address: 98.136.96.75
Name:   mta5.am0.yahoodns.net
Address: 67.195.204.79
Name:   mta5.am0.yahoodns.net
Address: 67.195.228.94
Name:   mta5.am0.yahoodns.net
Address: 98.136.96.77
Name:   mta5.am0.yahoodns.net
Address: 67.195.228.111
Name:   mta5.am0.yahoodns.net
Address: 67.195.228.110
Name:   mta5.am0.yahoodns.net
Address: 67.195.228.106
```

3. Tracing DNS with Wireshark

Part 3A:

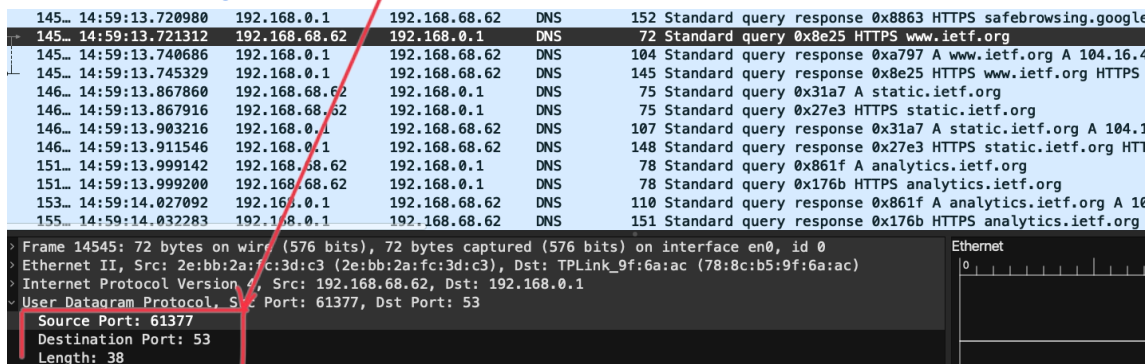
4. Locate the DNS query and response messages. Are then sent over UDP or TCP?

They are sent over UDP



5. What is the destination port for the DNS query message? What is the source port of DNS response message?

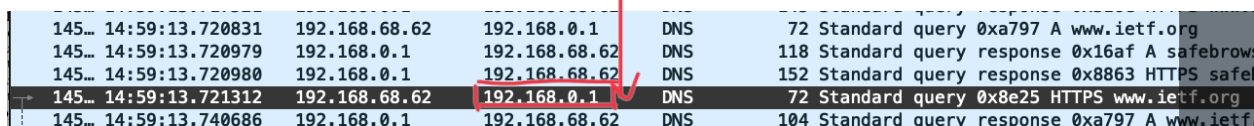
The destination port for a DNS query message is 53, and the source port of the DNS response message is 61377



6. To what IP address is the DNS query message sent? Use ipconfig to determine the IP address of your local DNS server. Are these two IP addresses the same?

Query message sent to IP address: 192.168.0.1.

The two IP addresses are the same.



```
> scutil --dns
DNS configuration

resolver #1
  nameserver[0] : 192.168.0.1
  nameserver[1] : 192.168.68.1
  if_index      : 11 (en0)
  flags         : Request A records
  reach         : 0x00000002 (Reachable)
```

7. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

It is a Type A query; the Queries message does not contain any answers.

```
3
Y
t
s
:
3
  v Queries
    v www.ietf.org: type A, class IN
      Name: www.ietf.org
      [Name Length: 12]
      [Label Count: 3]
      Type: A (1) (Host Address)
      Class: IN (0x0001)
```

8. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain?

2 answers are provided, and each answers contain Name, Type, Class, Time to live, Data length, and Address.

```

  Answers
  1  www.ietf.org: type A, class IN, addr 104.16.44.99
      Name: www.ietf.org
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 242 (4 minutes, 2 seconds)
      Data length: 4
      Address: 104.16.44.99
  2  www.ietf.org: type A, class IN, addr 104.16.45.99
      Name: www.ietf.org
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 242 (4 minutes, 2 seconds)
      Data length: 4
      Address: 104.16.45.99
      [Request In: 14542]
      [Time: 0.019855000 seconds]

```

9. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message?

The first SYN packet was sent to 104.16.45.99, which corresponds to the first IP address provided in the DNS response messages.

10. This web page contains images. Before retrieving each image, does your host issue new DNS queries? No

Part 3B: Do an *nslookup* on www.mit.edu

11. What is the destination port for the DNS query message? What is the source port of DNS response message?

Destination port: 53; Source port: 49236

12. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?

It was sent to the IP address 192.168.0.1; we can see from the screenshot below it is the default local DNS server.

ip.addr == 192.168.68.62						
No.	Time	Source	Destination	Protocol	Length	Info
209	15:33:25.938137	192.168.68.62	192.168.0.1	DNS	71	Standard query 0xb15b A www.mit.edu
210	15:33:25.966240	192.168.0.1	192.168.68.62	DNS	160	Standard query response 0xb15b A www.mit.e

```
> scutil --dns
DNS configuration

resolver #1
  nameserver[0] : 192.168.0.1
  nameserver[1] : 192.168.68.1
  if_index : 11 (en0)
  flags      : Request A records
  reach      : 0x00000002 (Reachable)
```

13. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

It is Type A, the query message doesn't contain any answers.

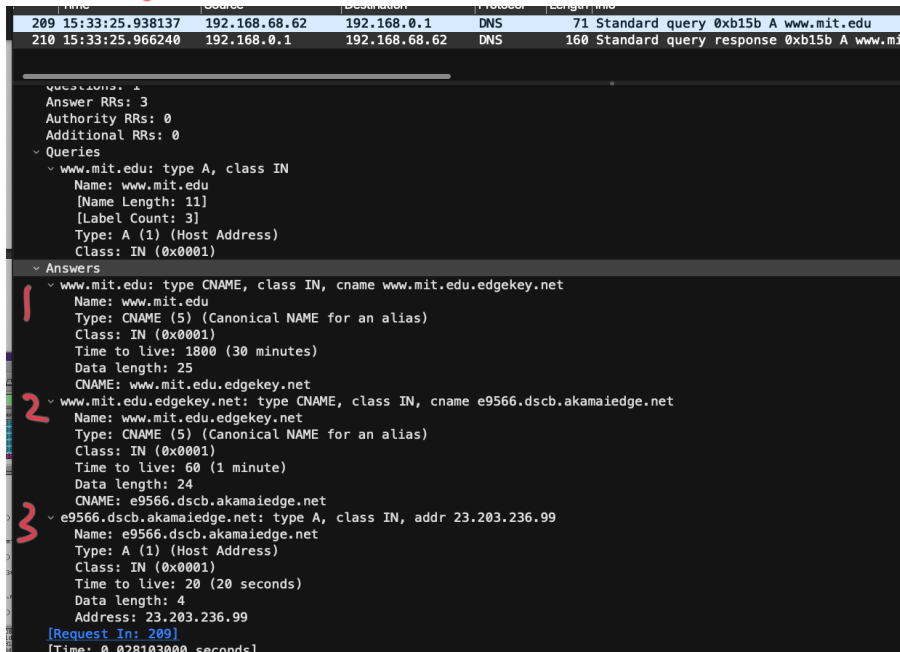
The image shows a Wireshark packet capture of a DNS query. The packet list at the top shows two packets: a standard query (No. 209) and a standard query response (No. 210). The details pane for packet 209 is expanded, showing the Domain Name System (query) section. Within this section, the 'Queries' list shows a query for 'www.mit.edu' with 'type A' and 'class IN'. A red box highlights the 'type A' field, and a red arrow points from the text 'It is Type A, the query message doesn't contain any answers.' to this field.

No.	Time	Source	Destination	Protocol	Length	Info
209	15:33:25.938137	192.168.68.62	192.168.0.1	DNS	71	Standard query 0xb15b A www.mit.edu
210	15:33:25.966240	192.168.0.1	192.168.68.62	DNS	160	Standard query response 0xb15b A www.n

[Checksum Status: Unverified]
[Stream index: 4]
[Stream Packet Number: 1]
> [Timestamps]
UDP payload (29 bytes)
~ Domain Name System (query)
Transaction ID: 0xb15b
> Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
~ Queries
~ www.mit.edu: type A, class IN
Name: www.mit.edu
[Name Length: 11]
[Label Count: 3]
Type: A (1) (Host Address)
Class: IN (0x0001)
[Response In: 210]

14. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain?

3 answers are provided, each answer contains: Name, Type, Class, Time to live, Data length, CNAME, and address.



```

Time      Source            Destination         Protocol  Length  Info
209 15:33:25.938137  192.168.68.62      192.168.0.1        DNS       71      Standard query 0xb15b A www.mit.edu
210 15:33:25.966240   192.168.0.1        192.168.68.62      DNS      160      Standard query response 0xb15b A www.m

Questions: 1
Answer RRs: 3
Authority RRs: 0
Additional RRs: 0
Queries
  www.mit.edu: type A, class IN
    Name: www.mit.edu
    [Name Length: 11]
    [Label Count: 3]
    Type: A (1) (Host Address)
    Class: IN (0x0001)
Answers
  www.mit.edu: type CNAME, class IN, cname www.mit.edu.edgekey.net
    Name: www.mit.edu
    Type: CNAME (5) (Canonical NAME for an alias)
    Class: IN (0x0001)
    Time to live: 1800 (30 minutes)
    Data length: 25
    CNAME: www.mit.edu.edgekey.net
  www.mit.edu.edgekey.net: type CNAME, class IN, cname e9566.dscb.akamaiedge.net
    Name: www.mit.edu.edgekey.net
    Type: CNAME (5) (Canonical NAME for an alias)
    Class: IN (0x0001)
    Time to live: 60 (1 minute)
    Data length: 24
    CNAME: e9566.dscb.akamaiedge.net
  e9566.dscb.akamaiedge.net: type A, class IN, addr 23.203.236.99
    Name: e9566.dscb.akamaiedge.net
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 20 (20 seconds)
    Data length: 4
    Address: 23.203.236.99
[Request In: 209]
[Time: 0.028103000 seconds]

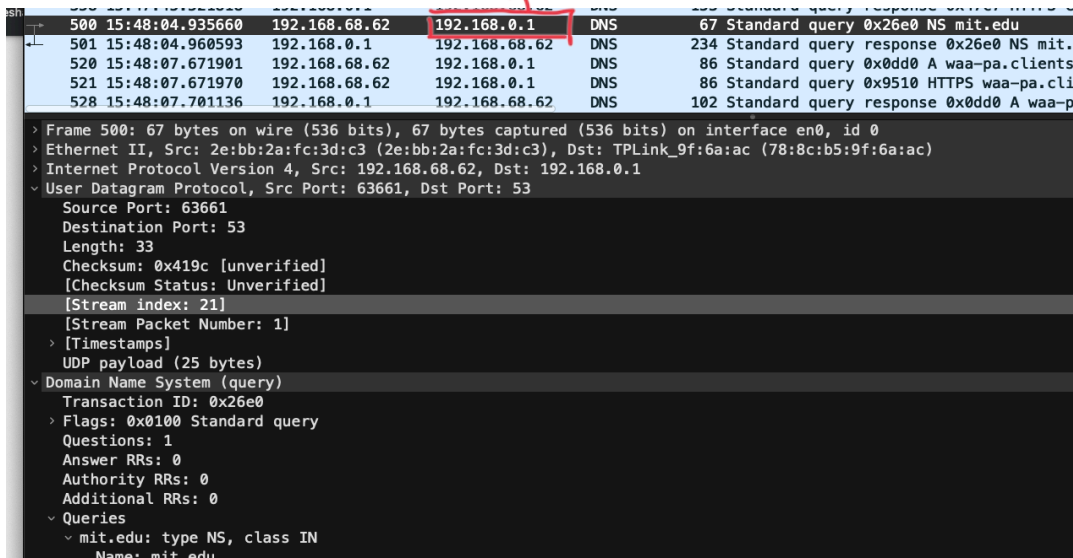
```

15. Provide a screenshot.

Part 3C: Do an nslookup -type=NS mit.edu

16. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?

It is sent to the IP address 192.168.0.1 which is my default server.



```

Time      Source            Destination         Protocol  Length  Info
500 15:48:04.935660   192.168.68.62      192.168.0.1        DNS       67      Standard query 0x26e0 NS mit.edu
501 15:48:04.960593   192.168.0.1        192.168.68.62      DNS      234      Standard query response 0x26e0 NS mit.
520 15:48:07.671901   192.168.68.62      192.168.0.1        DNS       86      Standard query 0x0dd0 A waa-pa.clients
521 15:48:07.671970   192.168.68.62      192.168.0.1        DNS       86      Standard query 0x9510 HTTPS waa-pa.cli
528 15:48:07.701136   192.168.0.1        192.168.68.62      DNS      102      Standard query response 0x0dd0 A waa-p

> Frame 500: 67 bytes on wire (536 bits), 67 bytes captured (536 bits) on interface en0, id 0
> Ethernet II, Src: 2e:bb:2a:fc:3d:c3 (2e:bb:2a:fc:3d:c3), Dst: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac)
> Internet Protocol Version 4, Src: 192.168.68.62, Dst: 192.168.0.1
> User Datagram Protocol, Src Port: 63661, Dst Port: 53
  Source Port: 63661
  Destination Port: 53
  Length: 33
  Checksum: 0x419c [unverified]
  [Checksum Status: Unverified]
  [Stream index: 21]
  [Stream Packet Number: 1]
  [Timestamps]
  UDP payload (25 bytes)
  Domain Name System (query)
    Transaction ID: 0x26e0
    Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
    Queries
      mit.edu: type NS, class IN
        Name: mit.edu

```

17. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

Type: NS, the query doesn't contain any answers

```

  ▾ Queries
    ▾ mit.edu: type NS, class IN
      Name: mit.edu
      [Name Length: 7]
      [Label Count: 2]
      Type: NS (2) (authoritative Name Server)
      Class: IN (0x0001)
      [Response In: 501]

```

18. Examine the DNS response message. What MIT nameservers does the response message provide? Does this response message also provide the IP addresses of the MIT nameservers?

It provided 8 nameservers:

- mit.edu nameserver = ns1-173.akam.net. 1
- mit.edu nameserver = usw2.akam.net. 2
- mit.edu nameserver = ns1-37.akam.net. 3
- mit.edu nameserver = use2.akam.net. 4
- mit.edu nameserver = eur5.akam.net. 5
- mit.edu nameserver = use5.akam.net. 6
- mit.edu nameserver = asia2.akam.net. 7
- mit.edu nameserver = asia1.akam.net. 8

Time	Source	Destination	Protocol	Length	Info
3	21:58:20.670655	192.168.68.64	192.168.0.1	DNS	74 Standard query 0xda97 A www.google.com
4	21:58:20.670795	192.168.68.64	192.168.0.1	DNS	74 Standard query 0x1dd9 HTTPS www.google.com
5	21:58:20.695394	192.168.0.1	192.168.68.64	DNS	90 Standard query response 0xda97 A www.google.com A 142.251.33.68
6	21:58:20.695397	192.168.0.1	192.168.68.64	DNS	99 Standard query response 0x1dd9 HTTPS www.google.com HTTPS
23	21:58:21.475308	192.168.68.64	192.168.0.1	DNS	90 Standard query 0x07a1 A config.extension.grammarly.com
24	21:58:21.475422	192.168.68.64	192.168.0.1	DNS	90 Standard query 0xb6a7 HTTPS config.extension.grammarly.com
25	21:58:21.495556	192.168.0.1	192.168.68.64	DNS	197 Standard query response 0x07a1 A config.extension.grammarly.com CNAME d27xe7juh1us6.cloudfront.n
26	21:58:21.502429	192.168.0.1	192.168.68.64	DNS	210 Standard query response 0xb6a7 HTTPS config.extension.grammarly.com CNAME d27xe7juh1us6.cloudfro
79	21:58:33.221955	192.168.68.64	192.168.0.1	DNS	67 Standard query 0x0273 NS mit.edu
80	21:58:33.241371	192.168.0.1	192.168.68.64	DNS	234 Standard query response 0x0273 NS mit.edu NS ns1-173.akam.net NS usw2.akam.net NS ns1-37.akam.net

▾ Queries

▾ mit.edu: type NS, class IN

▾ Answers

mit.edu: type NS, class IN, ns ns1-173.akam.net

Name: mit.edu

Type: NS (2) (authoritative Name Server)

Class: IN (0x0001)

Time to live: 1800 (30 minutes)

Data length: 18

Name Server: ns1-173.akam.net ✓ 1

mit.edu: type NS, class IN, ns usw2.akam.net

Name: mit.edu

Type: NS (2) (authoritative Name Server)

Class: IN (0x0001)

Time to live: 1800 (30 minutes)

Data length: 7

Name Server: usw2.akam.net ✓ 2

mit.edu: type NS, class IN, ns ns1-37.akam.net 3

Name: mit.edu

Type: NS (2) (authoritative Name Server)

Class: IN (0x0001)

Time to live: 1800 (30 minutes)

Data length: 9

Name Server: ns1-37.akam.net ✓

Ethernet

Destination	
Source	
Type	

Internet Protocol Version 4

Version	Header ...	Differentiated Services...	Total Length
Identification		Flags	Fragment Offset
Time to Live	Protocol	Header Checksum	
Source Address			
Destination Address			

User Datagram Protocol

--	--

80 21:58:33.241371 192.168.0.1 192.168.68.64 DNS 234 Standard query response 0x0273 NS mit.edu NS ns1-173.akam.net NS usw2.akam.net NS ns1-37.akam.net

```

~ mit.edu: type NS, class IN, ns use2.akam.net
  Name: mit.edu
  Type: NS (2) (authoritative Name Server)
  Class: IN (0x0001)
  Time to live: 1800 (30 minutes)
  Data length: 7
  Name Server: use2.akam.net
~ mit.edu: type NS, class IN, ns eur5.akam.net
  Name: mit.edu
  Type: NS (2) (authoritative Name Server)
  Class: IN (0x0001)
  Time to live: 1800 (30 minutes)
  Data length: 7
  Name Server: eur5.akam.net
~ mit.edu: type NS, class IN, ns use5.akam.net
  Name: mit.edu
  Type: NS (2) (authoritative Name Server)
  Class: IN (0x0001)
  Time to live: 1800 (30 minutes)
  Data length: 7
  Name Server: use5.akam.net
~ mit.edu: type NS, class IN, ns asia2.akam.net
  Name: mit.edu
  Type: NS (2) (authoritative Name Server)
  Class: IN (0x0001)
  Time to live: 1800 (30 minutes)
  Data length: 7
  Name Server: asia2.akam.net

```

4
✓ 5
✓ 6
✓ 7

Ethernet

Destination	
Source	
Type	

Internet Protocol Version 4

Version	Header ...	Differentiated Services...	Total Length
Identification		Flags	Fragment Offset
Time to Live	Protocol	Header Checksum	
Source Address		Destination Address	

User Datagram Protocol

Source Port	Destination Port
-------------	------------------

80 21:58:33.241371 192.168.0.1 192.168.68.64 DNS 234 Standard query response 0x0273 NS mit.edu NS ns1-173.akam.net NS usw2.akam.net NS ns1-37.akam.net

```

Data length: 7
Name Server: eur5.akam.net
~ mit.edu: type NS, class IN, ns use5.akam.net
  Name: mit.edu
  Type: NS (2) (authoritative Name Server)
  Class: IN (0x0001)
  Time to live: 1800 (30 minutes)
  Data length: 7
  Name Server: use5.akam.net
~ mit.edu: type NS, class IN, ns asia2.akam.net
  Name: mit.edu
  Type: NS (2) (authoritative Name Server)
  Class: IN (0x0001)
  Time to live: 1800 (30 minutes)
  Data length: 8
  Name Server: asia2.akam.net
~ mit.edu: type NS, class IN, ns asia1.akam.net
  Name: mit.edu
  Type: NS (2) (authoritative Name Server)
  Class: IN (0x0001)
  Time to live: 1800 (30 minutes)
  Data length: 8
  Name Server: asia1.akam.net
[Request ID: 79]
[Time: 0.019416000 seconds]

```

✓
✓ 8

Ethernet

Destination	
Source	
Type	

Internet Protocol Version 4

Version	Header ...	Differentiated Services...	Total Length
Identification		Flags	Fragment Offset
Time to Live	Protocol	Header Checksum	
Source Address		Destination Address	

User Datagram Protocol

Source Port	Destination Port
-------------	------------------

19. Provide a screenshot.

```

> nslookup -type=NS mit.edu
Server:          192.168.0.1
Address:         192.168.0.1#53

Non-authoritative answer:
mit.edu nameserver = ns1-173.akam.net.
mit.edu nameserver = usw2.akam.net.
mit.edu nameserver = ns1-37.akam.net.
mit.edu nameserver = use2.akam.net.
mit.edu nameserver = eur5.akam.net.
mit.edu nameserver = use5.akam.net.
mit.edu nameserver = asia2.akam.net.
mit.edu nameserver = asia1.akam.net.

Authoritative answers can be found from:

```

Part 3D: nslookup www.aiit.or.kr bitsy.mit.edu

20. To what IP address is the DNS query message sent? Is this the IP address of your

default local DNS server? If not, what does the IP address correspond to?

This DNS query message is sent to 192.168.0.1 which is the IP address of the MIT DNS response sender

No.	Time	Source	Destination	Protocol	Length	Info
27	21:30:06.682235	192.168.68.64	192.168.0.1	DNS	73	Standard query 0x3e6e A bitsy.mit.edu
28	21:30:06.704558	192.168.0.1	192.168.68.64	DNS	89	Standard query response 0x3e6e A bitsy.mit.edu A 18.0.72.3
31	21:30:06.708696	192.168.68.64	18.0.72.3	DNS	74	Standard query 0x290f A www.aiit.or.kr
92	21:30:11.709540	192.168.68.64	18.0.72.3	DNS	74	Standard query 0x290f A www.aiit.or.kr

21. Examine the DNS query message. What “Type” of DNS query is it? Does the Does query message contains any “answers”?

Type: Type A, the query message doesn't contain any answers

```
Additional RRs: 0
Queries
  bitsy.mit.edu: type A, class IN
    Name: bitsy.mit.edu
    [Name Length: 13]
    [Label Count: 3]
    Type: A (1) (Host Address)
    Class: IN (0x0001)
```

22. Examine the DNS response message. How many “answers” are provided? What does each of these answers contain?

One answer is provided, and the answer contains Name, type, Class, Time to live, Data length, and Address.

```
Class: IN (0x0001)
Answers
  bitsy.mit.edu: type A, class IN, addr 18.0.72.3
    Name: bitsy.mit.edu
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 197 (3 minutes, 17 seconds)
    Data length: 4
    Address: 18.0.72.3
[Request In: 27]
[Time: 0.022323000 seconds]
```

23. Provide a screenshot.

dns

No.	Time	Source	Destination	Protocol	Length	Info
27	21:30:06.682235	192.168.68.64	192.168.0.1	DNS	73	Standard query 0x3e6e A bitsy.mit.edu
28	21:30:06.704558	192.168.0.1	192.168.68.64	DNS	89	Standard query response 0x3e6e A bitsy.mit.edu A 18.0.72.3
31	21:30:06.708696	192.168.68.64	18.0.72.3	DNS	74	Standard query 0x290f A www.aiit.or.kr
92	21:30:11.709540	192.168.68.64	18.0.72.3	DNS	74	Standard query 0x290f A www.aiit.or.kr
104	21:30:14.358809	192.168.68.64	192.168.0.1	DNS	86	Standard query 0x7db8 A waa-pa.clients6.google.com
105	21:30:14.359043	192.168.68.64	192.168.0.1	DNS	86	Standard query 0x232c HTTPS waa-pa.clients6.google.com
112	21:30:14.383648	192.168.0.1	192.168.68.64	DNS	102	Standard query response 0x7db8 A waa-pa.clients6.google.com A 142.251.215.234
115	21:30:14.388569	192.168.0.1	192.168.68.64	DNS	136	Standard query response 0x232c HTTPS waa-pa.clients6.google.com 50A ns1.google.co
186	21:30:14.550487	192.168.68.64	192.168.0.1	DNS	90	Standard query 0x2f40 A config.extension.grammarly.com
187	21:30:14.550689	192.168.68.64	192.168.0.1	DNS	90	Standard query 0x814e HTTPS config.extension.grammarly.com
192	21:30:14.575932	192.168.0.1	192.168.68.64	DNS	210	Standard query response 0x814e HTTPS config.extension.grammarly.com CNAME d27xxe7

Answer RRs: 1
Authority RRs: 0
Additional RRs: 0

Queries

- bitsy.mit.edu: type A, class IN
 - Name: bitsy.mit.edu
 - [Name Length: 13]
 - [Label Count: 3]
 - Type: A (1) (Host Address)
 - Class: IN (0x0001)

Answers

- bitsy.mit.edu: type A, class IN, addr 18.0.72.3
 - Name: bitsy.mit.edu
 - Type: A (1) (Host Address)
 - Class: IN (0x0001)
 - Time to live: 197 (3 minutes, 17 seconds)
 - Data length: 4
 - Address: 18.0.72.3

[Request In: 27]
[Time: 0.022323000 seconds]

Ethernet

0151631

Destination

Source

Type

Internet Protocol Version 4

0151631

Version	Header ...	Differentiated Services...	Total Length	
Identification		Flags	Fragment Offset	
Time to Live	Protocol	Header Checksum		
Source Address				
Destination Address				

Destination Hardware Address (eth.dst): 6 bytes

Packets: 230 - Displayed: 12 (5.0%) - Dropped: 0 (0.0%)

Profile: Default